



A time-frequency-shift invariant parameter estimator for oscillating transient functions using the matched window reassignment

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ABSTRACT

In this paper we present the matched window reassignment method, generalizing the results to complex valued signals in multiple dimensions. For an oscillating transient signal with an envelope shape described by an arbitrary twice differentiable function, the reassigned spectrogram, with a matched window, concentrates all energy into one single time-frequency point. An estimator for the parameters of the envelope, in multiple dimensions, is constructed using the above property where the concentration is measured using the Rényi entropy. Furthermore, we present a classification scheme, where an observation is classified based on the concentration when reassigning with a set of model functions. Finally, two examples of parameter estimation from real-world measurements are shown, a one-dimensional time series of a single dolphin click and a two-dimensional time-series of seismic data.

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1. Introduction

Time-frequency representations (TFRs) are a set of important tools for analyzing non-stationary processes. Most time-frequency (TF) methods are based on linear or quadratic representations [1]. Among the linear transforms, the most well known are the short-time Fourier transform (STFT) and the wavelet transform. Other often applied linear transformations are the Gabor transform, a special case of the STFT [2], and the S-transform, a generalization of the STFT [3]. With specific restrictions, the linear transforms are invertible and allow for reconstruction of the signal.

The most applied quadratic (bi-linear) representations are the spectrogram and the scalogram, i.e., the quadratic transformations corresponding to the STFT and the wavelet transform [1]. The common ground for these and all other quadratic representations is the Wigner-Ville distribution (WVD), where the Hilbert transformed signal forms the instantaneous auto-correlation function that is Fourier transformed into a TF representation. The WVD has the best possible TF concentration for a single-component signal, but suffer severely from cross-terms for multi-component signals. Cross-terms occur also for all TF components generated from noise disturbances, which makes the WVD of noisy signals difficult to interpret and extract further information from. Any quadratic representation can be generated by applying a two-dimensional TF smoothing kernel to the WVD. The general aim of the kernel is to

keep the signal components and to suppress the cross-terms, and the research area of optimal and adapted TF smoothing kernels for different classes of signals and applications is extensive and well developed [4].

The Hilbert-Huang transform is the widely applied method to extract intrinsic modes, which often correspond to the instantaneous frequencies of the signal [5]. However, the method relies on empirical mode decomposition, where mode mixing is a general problem. Another important issue is that the number of components is usually unknown and thus need estimation [6].

The spectrogram is still a widely used TFR due to its fast implementation from the STFT and easy interpretation. It does however suffer from a trade-off in time and frequency resolution. The reassignment and synchrosqueezing are techniques designed to increase the resolution of the spectrogram [7–10]. These methods exploit the phase information, which is usually discarded in the quadratic distributions, and improves readability of the TFR. Several recent developments of these methods can be found, e.g., [11–16].

Transients occur in many acoustic measurements such as ultrasound analysis, seismic wave detection and vibration signal characterization. These transients are significant signatures in classification of epileptic seizures, and they are the key structures in animal communication, e.g., in signals from birds, bats and dolphins. It is often of interest to measure the above mentioned acoustic signals on a spatial grid, either on the surface of the Earth when detecting seismic waves, on an array of microphones to get information on the direction of animal sounds, or on the surface of a person's head when recording the electrical signals from the brain.

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The short duration of transients require that special care is taken when considering an analysis method. Short impulses are often modeled with a Gaussian envelope in time [17,18]. Modern algorithms that resolve the parameters of a Gaussian shaped function in time have been described by [19–21]. The Gabor and Stockwell transforms are widely applied TF methods. They are used and adapted in fields such as direction of arrival estimation [22], automatically adapting the TF resolution of transients [23], detecting epileptic seizures [24] and double-talk in acoustic echo cancellation [25]. Our scaled reassignment method, tailored for transients with Gaussian envelopes, has been shown to reach beyond the lower TF resolution bound of the Gabor transform [26]. We have further explored this technique for parameter estimation of oscillating transient Gaussian functions [27]. The method has further been applied for automatic component characterization of dolphin echolocation data [28–30].

However, for signals with components that are modulated in frequency and overlap heavily in time, assuming a Gaussian envelope might not give the best result. For this group of signals, TF based methods have been applied to a large extent, for which the main aim is to find the analysis window that achieves the best TF resolution [31]. We recently proposed a matched window reassignment (MWR), expanding the scaled reassignment method to transients with any envelope shape [32]. The method uses a window, matching the envelope, to find the TF centers of individual signal components and we have proved that perfect localization can be achieved for any signal with a twice differentiable and transient envelope function.

In this paper we further explore the MWR concept, generalizing the results to complex valued signals in multiple dimensions and exemplifying the MWR with a set of parameterized envelopes. The expansion to multiple dimensions will allow for better analysis of transients that have, e.g., spacial dependencies. The paper is structured as follows: in Section 2 we present the scaled reassigned spectrogram in n dimensions. In Section 3 we show that, if a matched window is used when calculating the reassignment vectors, perfect concentration into one single point may be achieved for any transient and twice differentiable function. In Section 4 we construct a parameter estimator and in Section 5, we show empirical results for the performance. Section 6 presents a classifier using the matched window reassignment property and Section 7 shows the proposed parameter estimator applied to an example of a recorded dolphin click and two-dimensional seismic data. Section 8 concludes the paper.

2. The scaled reassigned spectrogram in n -D

In this paper we are using properties of the reassigned spectrogram for parameter estimation and will allow for n -dimensional (n -D) signals. Even though the intermediate results will be visually difficult to present or interpret, the derived optimization criterion holds in any dimension. In deriving the MWR, we will assume that the dimensionality of the observed signal is arbitrarily large, but the examples shown in this paper will remain 1-D and 2-D.

Assume we have observations of a signal $x(\mathbf{t})$ that maps n real values onto a complex value, i.e.,

$$x(\mathbf{t}) : \mathbb{R}^n \rightarrow \mathbb{C}. \quad (1)$$

All multi-dimensional variables will be denoted as tuples, using bold symbols, where time is denoted

$$\mathbf{t} = (t_1, t_2, \dots, t_n), \quad (2)$$

and the corresponding frequency tuple as

$$\boldsymbol{\omega} = (\omega_1, \omega_2, \dots, \omega_n) \quad (3)$$

We will use τ as integrating variable in time and ν in frequency. Dot product is defined as

$$\mathbf{t} \cdot \boldsymbol{\omega} = \sum_{k=1}^n t_k \omega_k \quad (4)$$

Further, i will denote the imaginary unit, and $\Re\{\cdot\}$ and $\Im\{\cdot\}$ denote real and imaginary part, respectively. Complex conjugation will be denoted using the overline notation

$$\overline{\mathbf{a} + i\mathbf{b}} = \mathbf{a} - i\mathbf{b}. \quad (5)$$

The n -D Fourier transform is defined as, see e.g., [33],

$$X(\boldsymbol{\omega}) = \int_{\mathbb{R}^n} x(\boldsymbol{\tau}) e^{-i\boldsymbol{\omega} \cdot \boldsymbol{\tau}} d\boldsymbol{\tau} \quad (6)$$

and the inverse Fourier transform as

$$x(\mathbf{t}) = \int_{\mathbb{R}^n} X(\boldsymbol{\nu}) e^{i\boldsymbol{\nu} \cdot \mathbf{t}} \frac{d\boldsymbol{\nu}}{(2\pi)^n} \quad (7)$$

The short-time Fourier transform (STFT) of the signal $x(\mathbf{t})$ using window function $h(\mathbf{t})$ is defined as

$$F_x^h(\mathbf{t}, \boldsymbol{\omega}) = \int_{\mathbb{R}^n} x(\boldsymbol{\tau}) \overline{h(\boldsymbol{\tau} - \mathbf{t})} e^{-i\boldsymbol{\omega} \cdot \boldsymbol{\tau}} d\boldsymbol{\tau} \quad (8)$$

where a window function $h(\mathbf{t})$ is moved along the observed signal in every dimension, acting both as a smoothing function and a temporal window frame and the n -D Fourier transform is calculated at each position. The spectrogram is defined as the squared absolute value of the STFT, such that

$$S_x^h(\mathbf{t}, \boldsymbol{\omega}) = |F_x^h(\mathbf{t}, \boldsymbol{\omega})|^2. \quad (9)$$

A drawback of the spectrogram is the trade-off between resolution in time and in frequency. If a large temporal window is used, one will get a reliable estimate of the frequency content, but with poor information of how the frequency is distributed inside each wide temporal window. If a small temporal window function is used one will get a better time-resolution, but in turn get a worse frequency estimate. A proposed solution to this is the reassignment method, first proposed in [7], and thoroughly investigated in [8], where a sparse assumption is made, and energy in the TFR is moved to its local center of gravity. The reassignment may be expressed using the $2n$ -D Dirac delta function, $\delta(\cdot)$, as

$$RS_x^h(\mathbf{t}, \boldsymbol{\omega}) = \iint_{\mathbb{R}^{2n}} S_x^h(\boldsymbol{\tau}, \boldsymbol{\nu}) \delta(\mathbf{t} - \hat{\mathbf{t}}_x(\boldsymbol{\tau}, \boldsymbol{\nu}), \boldsymbol{\omega} - \hat{\boldsymbol{\omega}}_x(\boldsymbol{\tau}, \boldsymbol{\nu})) d\boldsymbol{\tau} \frac{d\boldsymbol{\nu}}{(2\pi)^n} \quad (10)$$

where

$$\hat{\mathbf{t}}_x(\mathbf{t}, \boldsymbol{\omega}) = (\hat{t}_x^{(1)}(\mathbf{t}, \boldsymbol{\omega}), \hat{t}_x^{(2)}(\mathbf{t}, \boldsymbol{\omega}), \dots, \hat{t}_x^{(n)}(\mathbf{t}, \boldsymbol{\omega})) \quad (11)$$

and

$$\hat{\boldsymbol{\omega}}_x(\mathbf{t}, \boldsymbol{\omega}) = (\hat{\omega}_x^{(1)}(\mathbf{t}, \boldsymbol{\omega}), \hat{\omega}_x^{(2)}(\mathbf{t}, \boldsymbol{\omega}), \dots, \hat{\omega}_x^{(n)}(\mathbf{t}, \boldsymbol{\omega})) \quad (12)$$

are the centers of gravity. A general definition of center of gravity, valid for TFRs of the quadratic class is

$$\hat{t}_x^{(j)}(\mathbf{t}, \boldsymbol{\omega}) = t_j - \frac{\iint_{\mathbb{R}^{2n}} \tau_j \phi(\boldsymbol{\tau}, \boldsymbol{\nu}) WV_x(\mathbf{t} - \boldsymbol{\tau}, \boldsymbol{\omega} - \boldsymbol{\nu}) d\boldsymbol{\tau} d\boldsymbol{\nu}}{\iint_{\mathbb{R}^{2n}} \phi(\boldsymbol{\tau}, \boldsymbol{\nu}) WV_x(\mathbf{t} - \boldsymbol{\tau}, \boldsymbol{\omega} - \boldsymbol{\nu}) d\boldsymbol{\tau} d\boldsymbol{\nu}} \quad (13)$$

$$\hat{\omega}_x^{(j)}(\mathbf{t}, \boldsymbol{\omega}) = \omega_j - \frac{\iint_{\mathbb{R}^{2n}} \nu_j \phi(\boldsymbol{\tau}, \boldsymbol{\nu}) WV_x(\mathbf{t} - \boldsymbol{\tau}, \boldsymbol{\omega} - \boldsymbol{\nu}) d\boldsymbol{\tau} d\boldsymbol{\nu}}{\iint_{\mathbb{R}^{2n}} \phi(\boldsymbol{\tau}, \boldsymbol{\nu}) WV_x(\mathbf{t} - \boldsymbol{\tau}, \boldsymbol{\omega} - \boldsymbol{\nu}) d\boldsymbol{\tau} d\boldsymbol{\nu}} \quad (14)$$

where $\phi(\mathbf{t}, \boldsymbol{\omega})$ is a chosen smoothing function and $WV_x(\mathbf{t}, \boldsymbol{\omega})$ is the WVD defined as

$$WV_x(\mathbf{t}, \boldsymbol{\omega}) = \int_{\mathbb{R}^n} x\left(\mathbf{t} + \frac{\boldsymbol{\tau}}{2}\right) \overline{x\left(\mathbf{t} - \frac{\boldsymbol{\tau}}{2}\right)} e^{-i\boldsymbol{\omega} \cdot \boldsymbol{\tau}} d\boldsymbol{\tau} \quad (15)$$

We are in this paper interested in reassignment of the spectrogram, and the center of gravity equations may then be expressed

Algorithm 1. Scaled reassigned spectrogram in n dimensions.

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- 1: Compute the STFTs $F_x^h[t, \omega]$, $F_x^{\mathcal{T}_j h}[t, \omega]$, and $F_x^{\mathcal{D}_j h}[t, \omega]$ for $j = 1, 2, \dots, n$ according to Eq. (8)
 - 2: Compute the spectrogram $S_x^h[t, \omega]$ according to Eq. (9)
 - 3: Compute the center of gravities $(\hat{t}_x(t, \omega), \hat{\omega}_x(t, \omega))$ according to Eqs. (18)-(19)
 - 4: Initiate $RS[t, \omega] = 0$
 - 5: **for** each point $[n, m]$ in $[t, \omega]$ **do**
 - 6: **if** $S_x^h[n, m] > \epsilon$ **then**
 - 7: $RS_x^h[\hat{t}_x[n, m], \hat{\omega}_x[n, m]] = S_x^h[n, m] + RS_x^h[\hat{t}_x[n, m], \hat{\omega}_x[n, m]]$
 - 8: **end if**
 - 9: **end for**
-

in a simplified format (see [8] for details)

$$\hat{t}_x^{(j)}(\mathbf{t}, \omega) = t_j + \Re \left\{ \frac{F_x^{\mathcal{T}_j h}(\mathbf{t}, \omega)}{F_x^h(\mathbf{t}, \omega)} \right\} \quad (16)$$

$$\hat{\omega}_x^{(j)}(\mathbf{t}, \omega) = \omega_j - \Im \left\{ \frac{F_x^{\mathcal{D}_j h}(\mathbf{t}, \omega)}{F_x^h(\mathbf{t}, \omega)} \right\}. \quad (17)$$

Here, the time-multiplication operator is $\mathcal{T}_j h = t_j h(\mathbf{t})$ and differentiation operator is $\mathcal{D}_j h = \partial h(\mathbf{t}) / \partial t_j$. Note that this means that for reassignment of a spectrogram of an n -D signal, one has to calculate $2n + 1$ number of STFTs.

If $F_x^h(\mathbf{t}, \omega) = 0$ for some value $(\mathbf{t}, \omega)$, the fractions in Eqs. (16) and (17) will be undefined, but as this also mean that $S_x^h(\mathbf{t}, \omega) = 0$, there is no mass to reassign. Hence, when calculating the reassigned spectrogram, we only calculate the reassignment vectors when $S_x^h(\mathbf{t}, \omega) > \epsilon$, for a chosen small, positive value ϵ .

As stated previously, the reassignment is derived with sparsity in mind. Given a sinusoidal function or an impulse, a perfect concentration is achieved. This is however not true if the sparsity assumption does not hold, i.e., the signal has a distribution in its TFR. In [26,27,34] we show, that perfect concentration may be achieved for the first two Hermite functions, if a matched window is used and we introduce a scaling factor in the reassignment. The reassignment vectors for the spectrogram Eqs. (16)-(17) are then redefined as

$$\hat{t}_x^{(j)}(\mathbf{t}, \omega) = t_j + c_t^{(j)} \Re \left\{ \frac{F_x^{\mathcal{T}_j h}(\mathbf{t}, \omega)}{F_x^h(\mathbf{t}, \omega)} \right\} \quad (18)$$

$$\hat{\omega}_x^{(j)}(\mathbf{t}, \omega) = \omega_j - c_\omega^{(j)} \Im \left\{ \frac{F_x^{\mathcal{D}_j h}(\mathbf{t}, \omega)}{F_x^h(\mathbf{t}, \omega)} \right\} \quad (19)$$

where scaling factors $c_t^{(j)}$ and $c_\omega^{(j)}$ are included. The algorithm for computing the scaled reassignment in n dimensions in the discrete TF case is presented in Algorithm 1.

3. The matched window reassignment

A transient oscillating signal is defined as

$$x(\mathbf{t}) = a(\mathbf{t} - \mathbf{t}_0) e^{i(\omega_0 \mathbf{t} + \phi)} \quad (20)$$

where $a(\mathbf{t})$ represents the transient envelope of the oscillation, $\mathbf{t}_0$ is a constant time-shift, ω_0 is a constant frequency shift, and ϕ is the phase. We note that we here will employ a broader definition of envelope, where we also include non-constant frequency shifts as part of the envelope function. The quadratic class and the reassignment operation are time-shift invariant [8]. This means that a translation in time and modulation in frequency of a signal only

results in a translation and modulation of its TFR. For the signal in Eq. (20), this gives that

$$RS_x^h(\mathbf{t}, \omega) = RS_a^h(\mathbf{t} - \mathbf{t}_0, \omega - \omega_0). \quad (21)$$

All further analysis can hence be restricted to $x(\mathbf{t}) = a(\mathbf{t})$, with time and frequency center $\mathbf{t}_0 = \omega_0 = \mathbf{0}$. By using the matched window

$$h(\mathbf{t}) = x(-\mathbf{t}) \triangleq x^- \quad (22)$$

we show that perfect localization may be achieved for any twice differentiable signal. To this end, we need to look at the scaled reassignment vectors, defined in Eqs. (18)-(19). As we are considering the case where the observed function is not time- or frequency shifted, we want all energy in the spectrogram to be reassigned to the origin, i.e., we want $\hat{t}_x^{(j)}(\mathbf{t}, \omega) = 0$ and $\hat{\omega}_x^{(j)}(\mathbf{t}, \omega) = 0$, $\forall j \in (1, \dots, n)$ and $\forall (\mathbf{t}, \omega) \in \mathbb{R}^{2n}$.

3.1. STFT With window x^-

We begin by calculating the STFT with the matched window

$$F_x^{x^-}(\mathbf{t}, \omega) = \int_{\mathbb{R}^n} x(\boldsymbol{\tau}) \overline{x^-(\boldsymbol{\tau} - \mathbf{t})} e^{-i\omega \cdot \boldsymbol{\tau}} d\boldsymbol{\tau} \quad (23)$$

$$= \int_{\mathbb{R}^n} x(\boldsymbol{\tau}) \overline{x(\mathbf{t} - \boldsymbol{\tau})} e^{-i\omega \cdot \boldsymbol{\tau}} d\boldsymbol{\tau} \quad (24)$$

Perform change of variable $\boldsymbol{\tau} = \boldsymbol{\tau}' + \frac{\mathbf{t}}{2}$. The Jacobian for this change of variable is the identity matrix, hence $d\boldsymbol{\tau} = d\boldsymbol{\tau}'$.

$$F_x^{x^-}(\mathbf{t}, \omega) = e^{-i\omega \cdot \frac{\mathbf{t}}{2}} \int_{\mathbb{R}^n} x\left(\boldsymbol{\tau}' + \frac{\mathbf{t}}{2}\right) \overline{x\left(-\boldsymbol{\tau}' + \frac{\mathbf{t}}{2}\right)} e^{-i\omega \cdot \boldsymbol{\tau}'} d\boldsymbol{\tau}' \quad (25)$$

$$\triangleq e^{-i\omega \cdot \frac{\mathbf{t}}{2}} R_1(\mathbf{t}, \omega) \quad (26)$$

Here $R_1(\mathbf{t}, \omega)$ is strictly real valued, as $R_1(\mathbf{t}, \omega) = \overline{R_1(\mathbf{t}, \omega)}$.

3.2. STFT With window $\mathcal{T}_j x^-$

We now turn to the STFTs using the time-shifted matched windows, needed to calculate the reassignment in each time dimension, as expressed in Eq. (16). The window function necessary for calculating the center of gravity in the j :th time dimension is then

$$\mathcal{T}_j h(\mathbf{t}) = t_j x(-\mathbf{t}) \quad (27)$$

The STFT is calculated as

$$F_x^{\mathcal{T}_j x^-}(\mathbf{t}, \omega) = \int_{\mathbb{R}^n} x(\boldsymbol{\tau}) (t_j - t_j) \overline{x(\mathbf{t} - \boldsymbol{\tau})} e^{-i\omega \cdot \boldsymbol{\tau}} d\boldsymbol{\tau} \quad (28)$$

$$= \int_{\mathbb{R}^n} x(\boldsymbol{\tau}) \left(t_j - \frac{t_j}{2} \right) \overline{x(\mathbf{t} - \boldsymbol{\tau})} e^{-i\omega \cdot \boldsymbol{\tau}} d\boldsymbol{\tau} - \frac{t_j}{2} F_x^{x^-}(\mathbf{t}, \omega) \quad (29)$$

Concentrate on the remaining integral, and perform change of variable $\boldsymbol{\tau} = \boldsymbol{\tau}' + \frac{\mathbf{t}}{2}$

$$F_x^{\mathcal{T}_j x^-}(\mathbf{t}, \omega) = e^{-i\omega \cdot \frac{\mathbf{t}}{2}} \int_{\mathbb{R}^n} t_j' x\left(\boldsymbol{\tau}' + \frac{\mathbf{t}}{2}\right) \overline{x\left(-\boldsymbol{\tau}' + \frac{\mathbf{t}}{2}\right)} e^{-i\omega \cdot \boldsymbol{\tau}'} d\boldsymbol{\tau}' - \frac{t_j}{2} F_x^{x^-}(\mathbf{t}, \omega) \quad (30)$$

$$\triangleq e^{-i\omega \cdot \frac{\mathbf{t}}{2}} \left(I_1^{(j)}(\mathbf{t}, \omega) - \frac{t_j}{2} R_1(\mathbf{t}, \omega) \right) \quad (31)$$

where $I_1^{(j)}(\mathbf{t}, \omega)$ is strictly imaginary valued, as $\overline{I_1^{(j)}(\mathbf{t}, \omega)} = -I_1^{(j)}(\mathbf{t}, \omega)$. Calculating the reassignment in time for the j :th dimension, defined in Eq. (18), we get

$$\hat{t}_x^{(j)}(\mathbf{t}, \omega) = t_j + c_t^{(j)} \Re \left\{ \frac{F_x^{\mathcal{T}_j h}(\mathbf{t}, \omega)}{F_x^h(\mathbf{t}, \omega)} \right\} \quad (32)$$

$$= t_j + c_t^{(j)} \Re \left\{ \frac{e^{-i\frac{\omega}{2}} \left(I_1^{(j)}(\mathbf{t}, \boldsymbol{\omega}) - \frac{t_j}{2} R_1(\mathbf{t}, \boldsymbol{\omega}) \right)}{e^{-i\frac{\omega}{2}} R_1(\mathbf{t}, \boldsymbol{\omega})} \right\} \quad (33)$$

$$= t_j - c_t^{(j)} \frac{t_j}{2} \quad (34)$$

By fixing $c_t^{(j)} = 2$, $\forall j = 1, \dots, n$, we see that

$$\hat{t}_x^{(j)}(\mathbf{t}, \boldsymbol{\omega}) = 0, \quad j = 1, \dots, n \quad (35)$$

for any $(\mathbf{t}, \boldsymbol{\omega}) \in \mathbb{R}^{2n}$

3.3. STFT With window $\mathcal{D}_j x^-$

We go on to calculate the STFTs for the spectral dimensions. Here the window function for finding the center of gravity in the j :th spectral dimension is

$$\mathcal{D}_j h(\mathbf{t}) = \frac{\partial x(-\mathbf{t})}{\partial t_j} \quad (36)$$

The STFT may then be calculated as

$$F_x^{\mathcal{D}_j x^-}(\mathbf{t}, \boldsymbol{\omega}) = - \int_{\mathbb{R}^n} x(\boldsymbol{\tau}) \overline{\frac{\partial x}{\partial t_j}(\mathbf{t} - \boldsymbol{\tau})} e^{-i\boldsymbol{\omega} \cdot \boldsymbol{\tau}} d\boldsymbol{\tau} \quad (37)$$

Transform integral to the spectral domain

$$F_x^{\mathcal{D}_j x^-}(\mathbf{t}, \boldsymbol{\omega}) = - \int_{\mathbb{R}^n} \left[\int_{\mathbb{R}^n} X(\mathbf{v}) e^{i\mathbf{v} \cdot \boldsymbol{\tau}} \frac{d\mathbf{v}}{(2\pi)^n} \right] \overline{\frac{\partial x}{\partial t_j}(\mathbf{t} - \boldsymbol{\tau})} e^{-i\boldsymbol{\omega} \cdot \boldsymbol{\tau}} d\boldsymbol{\tau} \quad (38)$$

$$= - \int_{\mathbb{R}^n} X(\mathbf{v}) \left[\int_{\mathbb{R}^n} \overline{\frac{\partial x}{\partial t_j}(\mathbf{t} - \boldsymbol{\tau})} e^{-i\boldsymbol{\omega} \cdot \boldsymbol{\tau}} e^{i\mathbf{v} \cdot \boldsymbol{\tau}} d\boldsymbol{\tau} \right] \frac{d\mathbf{v}}{(2\pi)^n} \quad (39)$$

$$= - \int_{\mathbb{R}^n} X(\mathbf{v}) i(v_j - \omega_j) \overline{X(-\mathbf{v} + \boldsymbol{\omega})} e^{i(\mathbf{v} - \boldsymbol{\omega}) \cdot \mathbf{t}} \frac{d\mathbf{v}}{(2\pi)^n} \quad (40)$$

Perform change of variable $\mathbf{v} = \mathbf{v}' + \frac{\boldsymbol{\omega}}{2}$, which gives

$$F_x^{\mathcal{D}_j x^-}(\mathbf{t}, \boldsymbol{\omega}) = e^{-i\frac{\boldsymbol{\omega} \cdot \mathbf{t}}{2}} \int_{\mathbb{R}^n} i \left(\frac{\omega_j}{2} - v'_j \right) X\left(\mathbf{v}' + \frac{\boldsymbol{\omega}}{2}\right) \overline{X\left(-\mathbf{v}' + \frac{\boldsymbol{\omega}}{2}\right)} e^{i\mathbf{v}' \cdot \mathbf{t}} \frac{d\mathbf{v}'}{(2\pi)^n} \quad (41)$$

The integral can be split into two parts, which we here denote

$$F_x^{\mathcal{D}_j x^-}(\mathbf{t}, \boldsymbol{\omega}) \triangleq e^{-i\frac{\boldsymbol{\omega} \cdot \mathbf{t}}{2}} i \left(\frac{\omega_j}{2} R_2(\mathbf{t}, \boldsymbol{\omega}) - I_2^{(j)}(\mathbf{t}, \boldsymbol{\omega}) \right) \quad (42)$$

Here, the second term is

$$I_2^{(j)}(\mathbf{t}, \boldsymbol{\omega}) = \int_{\mathbb{R}^n} v'_j X\left(\mathbf{v}' + \frac{\boldsymbol{\omega}}{2}\right) \overline{X\left(-\mathbf{v}' + \frac{\boldsymbol{\omega}}{2}\right)} e^{i\mathbf{v}' \cdot \mathbf{t}} \frac{d\mathbf{v}'}{(2\pi)^n} \quad (43)$$

which is purely imaginary as $\overline{I_2^{(j)}(\mathbf{t}, \boldsymbol{\omega})} = -I_2^{(j)}(\mathbf{t}, \boldsymbol{\omega})$. The first term is

$$R_2(\mathbf{t}, \boldsymbol{\omega}) = \int_{\mathbb{R}^n} X\left(\mathbf{v}' + \frac{\boldsymbol{\omega}}{2}\right) \overline{X\left(-\mathbf{v}' + \frac{\boldsymbol{\omega}}{2}\right)} e^{i\mathbf{v}' \cdot \mathbf{t}} \frac{d\mathbf{v}'}{(2\pi)^n} \quad (44)$$

and by transforming the integral back to the time domain

$$R_2(\mathbf{t}, \boldsymbol{\omega}) = \int_{\mathbb{R}^n} \left[\int_{\mathbb{R}^n} x(\boldsymbol{\tau}) e^{-i\frac{\boldsymbol{\omega} \cdot \boldsymbol{\tau}}{2}} e^{-i\mathbf{v}' \cdot \boldsymbol{\tau}} d\boldsymbol{\tau} \right] \overline{X\left(-\mathbf{v}' + \frac{\boldsymbol{\omega}}{2}\right)} e^{i\mathbf{v}' \cdot \mathbf{t}} \frac{d\mathbf{v}'}{(2\pi)^n} \quad (45)$$

$$= \int_{\mathbb{R}^n} x(\boldsymbol{\tau}) \left[\int_{\mathbb{R}^n} \overline{X\left(\mathbf{v}' + \frac{\boldsymbol{\omega}}{2}\right)} e^{-i\mathbf{v}' \cdot \boldsymbol{\tau}} e^{i\mathbf{v}' \cdot \mathbf{t}} \frac{d\mathbf{v}'}{(2\pi)^n} \right] e^{-i\frac{\boldsymbol{\omega} \cdot \boldsymbol{\tau}}{2}} d\boldsymbol{\tau} \quad (46)$$

$$= \int_{\mathbb{R}^n} x(\boldsymbol{\tau}) \overline{x(-\boldsymbol{\tau} + \mathbf{t})} e^{-i\boldsymbol{\omega} \cdot (\boldsymbol{\tau} - \frac{\mathbf{t}}{2})} d\boldsymbol{\tau} \quad (47)$$

and perform the change of variable $\boldsymbol{\tau} = \boldsymbol{\tau}' + \frac{\mathbf{t}}{2}$ we find that

$$R_2(\mathbf{t}, \boldsymbol{\omega}) = \int_{\mathbb{R}^n} x\left(\boldsymbol{\tau}' + \frac{\mathbf{t}}{2}\right) \overline{x(-\boldsymbol{\tau}' + \frac{\mathbf{t}}{2})} e^{-i\boldsymbol{\omega} \cdot \boldsymbol{\tau}'} d\boldsymbol{\tau}' \quad (48)$$

$$= R_1(\mathbf{t}, \boldsymbol{\omega}) \quad (49)$$

By inserting the results into Eq. (19), we get the reassignment in frequency as

$$\hat{\omega}_x^{(j)}(\mathbf{t}, \boldsymbol{\omega}) = \omega_j - c_\omega^{(j)} \Im \left\{ \frac{F_x^{\mathcal{D}_j h}(\mathbf{t}, \boldsymbol{\omega})}{F_x^h(\mathbf{t}, \boldsymbol{\omega})} \right\} \quad (50)$$

$$= \omega_j - c_\omega^{(j)} \Im \left\{ \frac{e^{-i\frac{\omega}{2}} i \left(\frac{\omega_j}{2} R_1(\mathbf{t}, \boldsymbol{\omega}) - I_2^{(j)}(\mathbf{t}, \boldsymbol{\omega}) \right)}{e^{-i\frac{\omega}{2}} R_1(\mathbf{t}, \boldsymbol{\omega})} \right\} \quad (51)$$

$$= \omega_j - c_\omega^{(j)} \frac{\omega_j}{2} \quad (52)$$

By fixing $c_\omega^{(j)} = 2$, $\forall j = 1, \dots, n$, we see that

$$\hat{\omega}_x^{(j)}(\mathbf{t}, \boldsymbol{\omega}) = 0, \quad \forall j = 1, \dots, n \quad (53)$$

and for any $(\mathbf{t}, \boldsymbol{\omega}) \in \mathbb{R}^{2n}$

3.4. Perfect concentration of the matched window reassignment

Studying the results from Eqs. (35) and (53), both are identically zero, independently of the time and frequency location in the TF-domain. This means that every point will be reassigned to the origin, giving a perfect concentration with all energy of the TFR located in a single point. As the reassignment in each dimension is described by a ratio of two STFTs, where the window function in the numerator is a linear operator applied to the window function of the denominator, perfect concentration is retained when the matched window function is scaled by a multiplicative constant. Furthermore, as the STFT is time and frequency-shift invariant, a signal shifted t_0 seconds in time and ω_0 rad/s in frequency, will give a MWR with all energy at the point corresponding to the shift. The reassignment vectors will then be

$$\hat{t}_x^{(j)}(\mathbf{t}, \boldsymbol{\omega}) = t_0^{(j)}, \quad \forall j = 1, \dots, n \quad (54)$$

$$\hat{\omega}_x^{(j)}(\mathbf{t}, \boldsymbol{\omega}) = \omega_0^{(j)}, \quad \forall j = 1, \dots, n \quad (55)$$

In theory, this will hold for any functions such that the integrals calculating the STFTs exist. In practice, we need it to be transient as we need to observe, and create a matched window function, covering the majority of the energy of the signal. We also need the derivative of the function to be smooth, such that the window functions $\mathcal{D}_j h$ are approximately bandlimited. To illustrate how the MWR of the spectrogram works in practice, we will show two examples; the first one in 1-D and the second one in 2-D.

3.5. An example in 1-D

We will start by going through the necessary steps for a 1-D function, to clearly exemplify and visualize the steps of the proposed algorithm. As a simulation model we will use a Cauchy function

$$a(t, \theta) = \frac{1}{1 + \left(\frac{t}{\theta}\right)^2} \quad (56)$$

where the observed function may be given time and frequency shift, denoted t_0 and ω_0 , as well as amplitude and phase shift, denoted α and ϕ

$$x(t, \theta) = \alpha a(t - t_0) e^{i(\omega_0 t + \phi)} \quad (57)$$

An example of a realization is shown in Fig. 4a, showing the real part of the function with parameters $\theta = 5\sqrt{2}$, $t_0 = -10$ s, $\omega_0 = \pi/3$ rad/s. Next we define the matched window functions

$$h(\theta) = \frac{1}{1 + \left(\frac{t}{\theta}\right)^2} \quad (58)$$

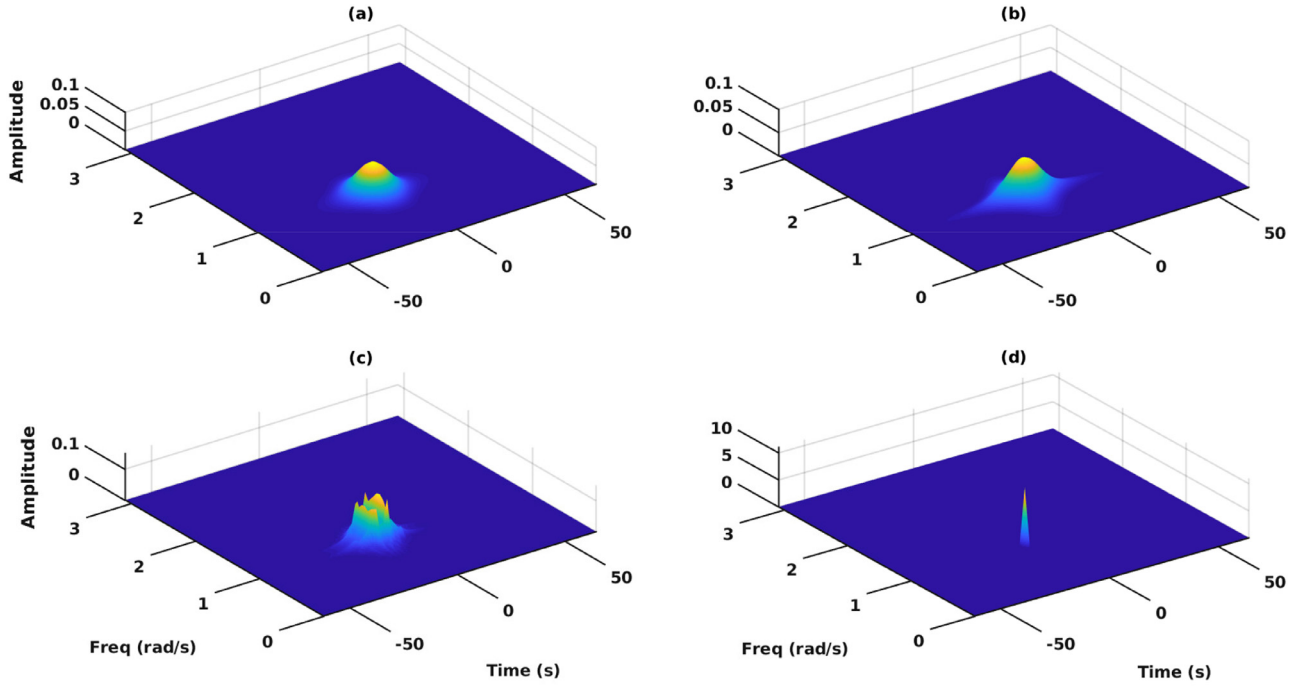


Fig. 1. Figure (a) shows the spectrogram of a Cauchy function using a Hann window. Figure (b) shows the matched window spectrogram. Figure (c) depicts the reassigned spectrogram, when using a Hann window. Figure (d) is the matched window reassigned spectrogram of the Cauchy function. When comparing figures (c) and (d), note especially the range of the z-axis, reflecting that all energy of the spectrogram is moved to one single point.

$$\mathcal{T}h(\theta) = \frac{t}{1 + \left(\frac{-t}{\theta}\right)^2} \quad (59)$$

$$\mathcal{D}h(\theta) = \frac{-2t}{\theta^2} \frac{1}{\left(1 + \left(\frac{-t}{\theta}\right)^2\right)^2} \quad (60)$$

Here, as the function is 1-D we have dropped the indexing of the time-shift- and differential operators. Following [Algorithm 1](#) one can now, using these three window functions, compute the necessary STFTs according to [Eq. \(8\)](#), one for each window function. The spectrogram is calculated as [Eq. \(9\)](#) and the reassignment vectors as [Eqs. \(18\)-\(19\)](#). The resulting spectrogram, and MWR spectrogram are shown in [Fig. 1b](#) and [1 d](#). As comparison we calculate the spectrogram using a Hann window [1] of length 32 and apply the original reassignment method [8], and depict the resulting TFRs in [Fig. 1a](#) and [1 c](#). Note that perfect concentration of the mass into one single point is achieved for the MWR, with a high energy peak at the localization point.

3.6. An example in 2-D

We go on to show a case where the signal is 2-D. The example signal here consist of a product of two Gaussian components, each with its own shape parameter, making the envelope function

$$a(\mathbf{t}, \boldsymbol{\theta}) = e^{-\frac{t_1^2}{2\theta_1^2}} e^{-\frac{t_2^2}{2\theta_2^2}} \quad (61)$$

with $\theta_1 = \theta_2 = 5\sqrt{2}$. Furthermore, the function is shifted -5 s in t_1 and 5 s in t_2 and given a frequency shift of $\pi/4$ rad/s in ω_1 and $-\pi/3$ rad/s in ω_2 and random phase shifts ϕ_1 and ϕ_2 , such that

$$x(\mathbf{t}, \boldsymbol{\theta}) = e^{-\frac{(t_1+5)^2}{2\theta_1^2}} e^{-\frac{(t_2-5)^2}{2\theta_2^2}} e^{i(\frac{\pi}{4}t_1+\phi_1)} e^{i(-\frac{\pi}{3}t_2+\phi_2)} \quad (62)$$

The real part of a realization of the signal is depicted in [Fig. 2](#). In [Fig. 3a-b](#) we show the time and frequency marginals of the spectrogram, respectively, when using the matched window function of

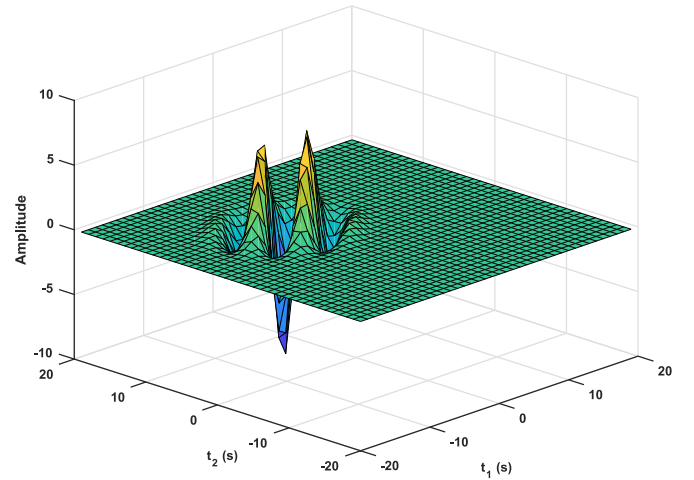


Fig. 2. The figure depicts the real part of a realization of a time and frequency shifted 2-D Gaussian specified in [Section 3.6](#), [Eq. \(62\)](#).

(61). As the function is 2-D, the spectrogram is 4-D, and hence we only depict the marginals. [Figs. 3c-d](#) show the time and frequency marginals after MWR, and the marginals correctly consist of one Dirac delta function each, i.e., all energy is reassigned to one single point. Note also that the mass is concentrated at the time and frequency location corresponding to the time and frequency shifts defined above.

4. Parameter estimation

In this section we propose a method for estimating the unknown parameter vector $\boldsymbol{\theta}$ given an observation of some parametric function $x(\mathbf{t}, \boldsymbol{\theta})$. This method uses the perfect concentration of the MWR, which is achieved when the window function matches the envelope function exactly, i.e., $h(\mathbf{t}, \boldsymbol{\theta}) = x(-\mathbf{t}, \boldsymbol{\theta})$. Here, the

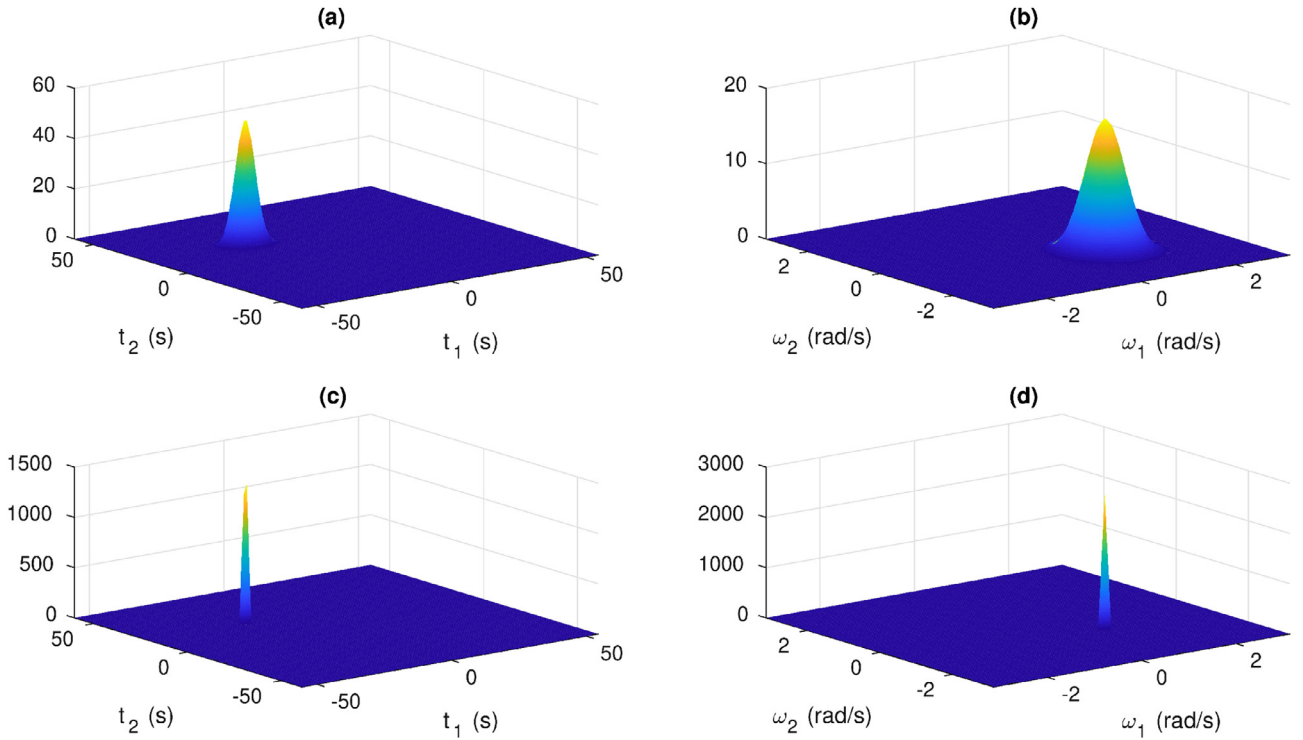


Fig. 3. The figure shows an example of the marginals of the spectrogram and the reassigned spectrogram of a 2-D Gaussian bell function. Figures (a) and (b) show the time and frequency marginal of the spectrogram, respectively. Figures (c) and (d) show the time and frequency marginal of the matched window reassigned spectrogram, respectively. Note that all energy has been reassigned to one single point, corresponding to the time and frequency shift of the function.

parameter vector $\theta = [\theta_1, \dots, \theta_m]$ may have any positive integer dimensionality, i.e., it may be independent of the dimensionality of the signal.

By defining the window function as matching the envelope function of the signal, but with unit energy, such that

$$h(\mathbf{q}) = \frac{1}{\sqrt{e_a}} a(-\mathbf{t}, \mathbf{q}) \quad (63)$$

we can tune $\mathbf{q}$ and find the parameter rendering the best concentration after reassignment. The energy of the envelope is

$$e_a = \int_{\mathbb{R}^n} |a(-\mathbf{t}, \mathbf{q})|^2 d\mathbf{t} \quad (64)$$

and by using a unit energy window function we ensure that the energy of the spectrogram stays constant for every window function, making the entropies of two spectrograms, calculated using two different parameters, comparable. To measure the concentration we propose that the Rényi entropy is used, in line with [27]. By denoting the reassigned spectrogram of an observation $x(\mathbf{t}, \theta)$ using window function $h(\mathbf{q})$ as $RS_x^{h(\mathbf{q})}$, the Rényi entropy is defined as

$$R\{RS_x^{h(\mathbf{q})}\} = \frac{1}{1-\beta} \log_2 \iint_{\mathbb{R}^{2n}} \left(RS_x^{h(\mathbf{q})}(\mathbf{t}, \omega) \right)^\beta d\mathbf{t} \frac{d\omega}{(2\pi)^n} \quad (65)$$

with $\beta = 3$, as per discussion in [35,36]. A parameter estimator could then be expressed as

$$\hat{\theta} = \underset{\mathbf{q}}{\operatorname{argmin}} R\{RS_x^{h(\mathbf{q})}\} \quad (66)$$

If the observed function $x(\mathbf{t}, \theta)$ is continuous in both $\mathbf{t}$ and θ , the local center-of-masses will move continuously with changes in θ . Hence, also the Rényi entropy of the reassigned spectrogram will change continuously. Further, we have empirically seen that a unimodal function (in both $\mathbf{t}$ and θ) renders a quasi-convex entropy curve. How a specific function behaves is easy to investigate when a model function is chosen.

Algorithm 2. Parameter estimation using matched window reassignment.

- 1: Define the parametric signal model $a(\mathbf{t}, \theta)$
 - 2: Define a dictionary of parameters $\mathbf{Q}_0$
 - 3: **for** $k = 0, \dots$, until convergence **do**
 - 4: **for** each $\mathbf{q}_k$ in $\mathbf{Q}_k$ **do**
 - 5: Calculate the STFTs $F_y^{h(\mathbf{q}_k)}$, $F_y^{T_j h(\mathbf{q}_k)}$ and $F_y^{D_j h(\mathbf{q}_k)}$, $\forall j = 1, \dots, n$
 - 6: Compute the reassigned spectrogram $RS_y^{h(\mathbf{q}_k)}$
 - 7: Compute the Rényi entropy $R_\alpha(RS_y^{h(\mathbf{q}_k)})$
 - 8: **end for**
 - 9: $\hat{\theta}_k = \underset{\mathbf{q} \in \mathbf{Q}_k}{\operatorname{argmin}} R_\alpha(RS_y^{h(\mathbf{q})})$
 - 10: Create zoomed in dictionary $\mathbf{Q}_{k+1}$ around $\hat{\theta}_k$
 - 11: **end for**
-

However, if the observed signals is disturbed by additive noise the quasi-convexity may no longer hold. We therefore propose a dictionary based algorithm, in line with the proposed algorithm in [27]. There we start by creating a coarsely spaced dictionary. The search for minima are further explored by creating zoomed-in dictionaries to increase the resolution. This may be repeated until one has achieved a satisfactory resolution

$$R_\alpha(RS_y^{h(\mathbf{q}_k)}) < \epsilon_r \quad (67)$$

or until convergence, i.e.,

$$\left| R_\alpha(RS_y^{h(\mathbf{q}_k)}) - R_\alpha(RS_y^{h(\mathbf{q}_{k-1})}) \right| < \epsilon_c \quad (68)$$

for some set ϵ_r or ϵ_c . The algorithm is summarized in Algorithm 2.

A non-linear least squares search may additionally be used to increase the accuracy of the estimated parameters, as was done in [27]. The MWR method may then be seen as an initial parameter estimator, where the time and frequency shift can be ignored. However, this is not included in the scope of this paper.

Assume that the observed signal is $N_1 \times N_2 \times \dots \times N_n$ samples and the discrete Fourier transform is calculated to M_ℓ frequency points in the ℓ :th dimension. The fast Fourier transform is then of computational complexity

$$\mathcal{O} \left\{ \left(\prod_{k=1}^n M_k \right) \log \left(\prod_{\ell=1}^n M_\ell \right) \right\}. \quad (69)$$

The computational complexity of both the reassignment and the Rényi entropy is

$$\mathcal{O} \left\{ \prod_{k=1}^n M_k \prod_{j=1}^n N_j \right\}. \quad (70)$$

If the dictionary contains K_μ proposal elements of the μ :th parameter, the total complexity of the estimator is

$$\mathcal{O} \left\{ \prod_{\mu=1}^m K_\mu \prod_{k=1}^n M_k \prod_{j=1}^n N_j \sum_{\ell=1}^n \log M_\ell \right\}. \quad (71)$$

If the underlying Rényi function is quasi-convex, the computational complexity of the parameter estimation may be improved by estimating each parameter cyclically. The complexity then reduces to

$$\mathcal{O} \left\{ K \prod_{k=1}^n M_k \prod_{j=1}^n N_j \sum_{\ell=1}^n \log M_\ell \right\} \quad (72)$$

where $K = \max(K_1, \dots, K_\mu)$.

5. Simulations

In this sections we will give examples of a few more or less commonly used models. We will here define a signal without noise as $x(\mathbf{t})$ where

$$x(\mathbf{t}) = \alpha a(\mathbf{t} - \mathbf{t}_0) e^{i(\omega_0 \mathbf{t} + \phi)} \quad (73)$$

Here $a(\mathbf{t})$ is the envelope of the signal, α regulates the amplitude, $\mathbf{t}_0$ is a time shift, ω_0 a frequency shift, and ϕ the phase. An observed signal with additive white spherically symmetric noise is denoted

$$y(\mathbf{t}) = x(\mathbf{t}) + z(\mathbf{t}) \quad (74)$$

We assume the covariance of the noise term, $z(\mathbf{t})$, to be $\sigma^2 I$, where I is the identity matrix. Further the signal-to-noise ratio (SNR_{dB}) is defined as

$$SNR_{dB} = 10 \log_{10} \left(\frac{P_x}{\sigma^2} \right) \quad (75)$$

where $P_x = \frac{1}{N_s} \sum_{\mathbf{t}} |x(\mathbf{t})|^2$ is the average power of the signal, and N_s the number of observed samples.

5.1. Cauchy function

We begin by applying the parameter estimator to the 1-D Cauchy function given as example in Section 3.5 and Eq. (56), also retaining the parameters chosen therein. An example of a realization is shown in Fig. 4a, showing the real part of the function with parameters $\theta = 5\sqrt{2}$, $t_0 = -10$ s, $\omega_0 = \pi/3$ rad/s. In this example, we assume that we know the parameter to be in the range $[2, 20]$, and we construct an equidistant grid of proposal parameters in this range. For each parameter in the dictionary we calculate the reassignment vectors according to Eqs. (18)-(19) and compute the reassigned spectrogram. We finally compute the Rényi entropy of each reassigned spectrogram according to Eq. (65). The resulting curve is seen in Fig. 4b. Note that the minimum of the Rényi curve coincides with the true value, $\theta = 5\sqrt{2}$, which is depicted as a red

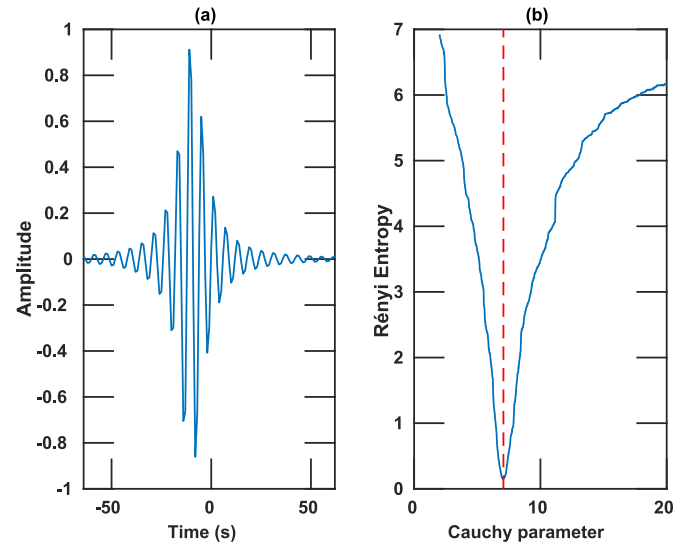


Fig. 4. Figure (a) shows a realization of the Cauchy function. In Figure (b) the Rényi entropy is shown as function of proposal parameters. The true parameter of the observed function is shown as a red vertical dashed line. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

vertical dashed line in the figure. The true parameter value is chosen as an irrational value to avoid on-grid hits with the equidistantly created proposal parameters in the dictionary.

To exemplify how the estimator performs for different levels of SNR_{dB} , we generate 250 observations each for the SNR_{dB} levels 0, 5, 10, ..., 30. Each realization is 150 samples long and with fixed amplitude $\alpha = 1$ and we draw a random parameters according to $\theta \in \mathcal{U}(2, 5)$, $t_0 \in \mathcal{U}(-5, 5)$, $\omega_0 \in \mathcal{U}(\pi/2 - 0.5, \pi/2)$ and $\phi \in \mathcal{U}(0, 2\pi)$. For each realization we estimate the parameter θ and we additionally calculate the Cramér-Rao Lower bound (CRLB). The dictionary is created containing 50 elements, and we zoom in 5 times, where the range of the dictionary is halved each time. The root-mean-squared error (RMSE), for the parameter θ defined as

$$RMSE = \sqrt{\frac{1}{N} \sum_{\eta=1}^N (\hat{\theta}_\eta - \theta_\eta)^2} \quad (76)$$

and the square-root of the mean CRLB for each SNR_{dB} level is depicted in Fig. 5. For high values of SNR, the proposed method performs well, rendering RMSE close to the possible lower bound. When the SNR decreases, the noise influences the reassignment vectors more, making the concentration, and consequently the parameter estimates, worse.

5.2. Gaussian function

We have extensively investigated the case of Gaussian function estimation in [27], but we will include a short description here to be used as a further example in this paper. The Gaussian envelope function is defined as

$$a(t, \theta) = e^{-\frac{t^2}{2\theta^2}} \quad (77)$$

and we generate observations of the Gaussian envelope with random parameters, $\theta \in \mathcal{U}(5, 10)$, time shift $t_0 \in \mathcal{U}(-5, 5)$, frequency shift $\omega_0 \in \mathcal{U}(\pi/2 - 0.5, \pi/2)$ and phase $\phi \in \mathcal{U}(0, 2\pi)$. We generate 250 observations each for the SNR_{dB} 0, 5, ..., 30, and compare with the CRLB and also with Guo's method applied to the Hilbert transform envelope of the observed signal [19]. The results are

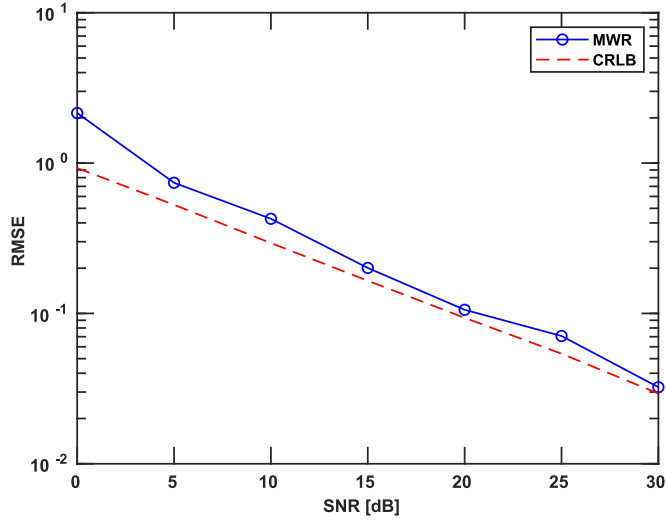


Fig. 5. The figure shows the RMSE of the Cauchy function shape parameter θ as function of the SNR_{dB} , estimated with the MWR method, compared to the CRLB. The performance of the proposed MWR method is close to the CRLB for $SNR_{dB} \gtrsim 15$.

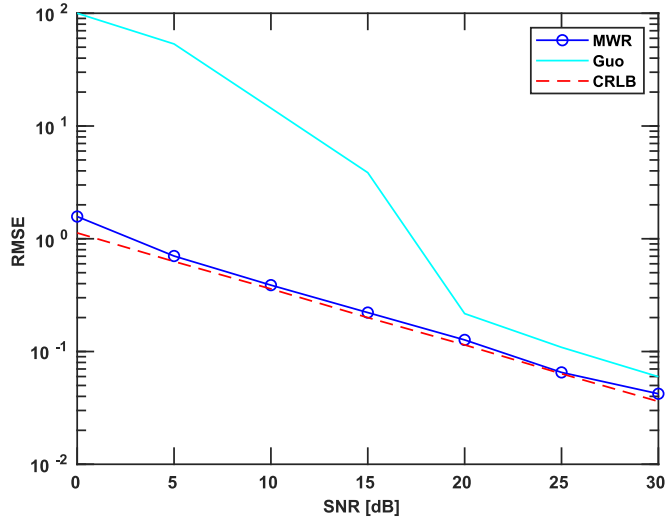


Fig. 6. The figure shows the RMSE of the Gaussian shape parameter θ as function of SNR_{dB} , estimated with the MWR method and compared to Guo's method applied to the envelope of the signal, and to the CRLB. The performance of the proposed MWR method is close to CRLB for $SNR_{dB} \gtrsim 5$.

shown in Fig. 6, where it is clearly seen that the MWR RMSE is close to the CRLB, also for low SNR_{dB} . Guo's method is not designed to handle frequency-modulated Gaussian signals and the envelope approximation of the noisy signal deteriorates with lowered SNR_{dB} .

5.3. Chirped transients

In this example we explore a 1-D signal model with two parameters. By setting $\theta = (\theta_1, \theta_2)$, we define the linearly chirped Gaussian function as

$$a(t, \theta) = e^{-\frac{t^2}{2\theta_1}} e^{j\frac{\theta_2 t^2}{2}} \quad (78)$$

meaning the Gaussian function is given an angle in the TFR. We created a grid of proposal parameter values and calculated the window function for each proposal parameter combination. The Rényi entropy was calculated for each reassignment performed for

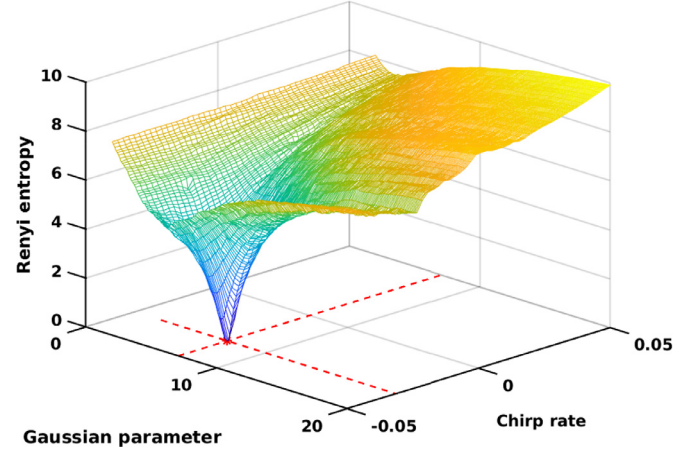


Fig. 7. The figure shows the Rényi entropy as function of the proposal parameters (θ_1, θ_2) of a chirped Gaussian function. The minimum matches the true parameter setting shown by the dashed red lines. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

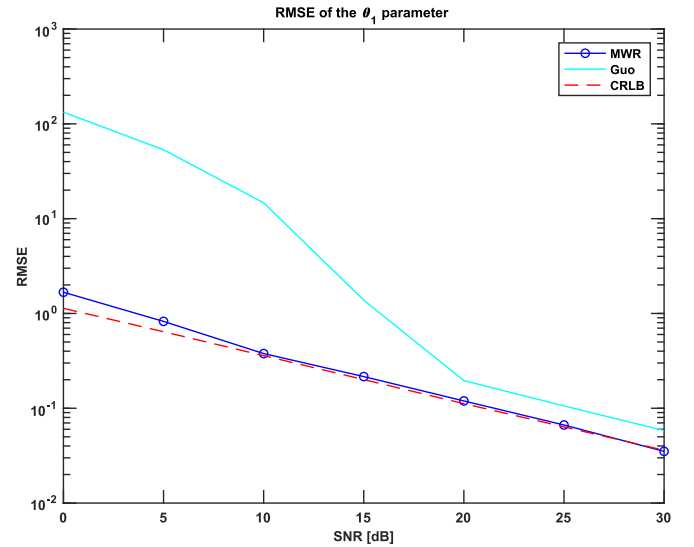


Fig. 8. The figure shows the RMSE of the estimation of the shape parameter θ_1 of a chirped Gaussian function. The parameter is estimated with the MWR method and compared to Guo's method applied to the envelope of the signal and the CRLB. The performance of the proposed MWR method is close to CRLB for $SNR_{dB} \gtrsim 10$.

each proposal parameter combination. The resulting Rényi surface is depicted in Fig. 7. The true parameter values (Gaussian parameter, chirp rate) are set to $\theta = (5\sqrt{2}, 0.01\pi)$. These values are shown as dashed red lines in Fig. 7.

We go on to exemplify the performance for a range of different SNR_{dB} . Here we generate observations for the same range of SNR_{dB} as in previous cases, with 250 realizations for each level of SNR_{dB} . We draw the random parameters of each realization such that Gaussian shape parameter $\theta_1 \in \mathcal{U}(5, 10)$, chirp rate $\theta_2 \in \mathcal{U}(-0.01, 0.01)$, time-shift $t_0 \in \mathcal{U}(-5, 5)$, frequency shift $\omega_0 \in \mathcal{U}(\pi/2 - 0.5, \pi/2)$ and phase $\phi \in \mathcal{U}(0, 2\pi)$. The Gaussian parameter θ_1 is compared with CRLB and Guo's method, whereas the chirp rate θ_2 is compared with CRLB, shown in Figs. 8 and 9, respectively. The results in Fig. 8 is similar to Fig. 6, and it is clearly seen that Guo's method is not appropriate for chirped Gaussian functions and the envelope approximation of the noisy signal deteriorates with lowered SNR_{dB} .

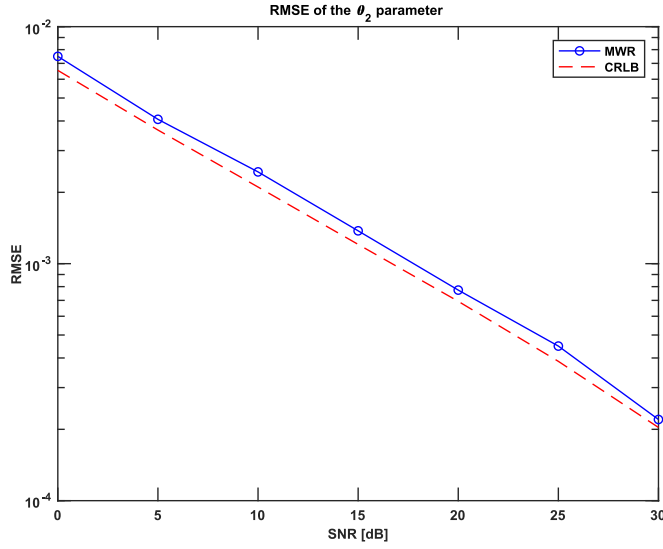


Fig. 9. The figure shows the RMSE of the chirp rate parameter θ_2 , estimated with the MWR method, compared to CRLB. The performance of the proposed MWR method is close to CRLB for the evaluated levels of SNR_{dB} .

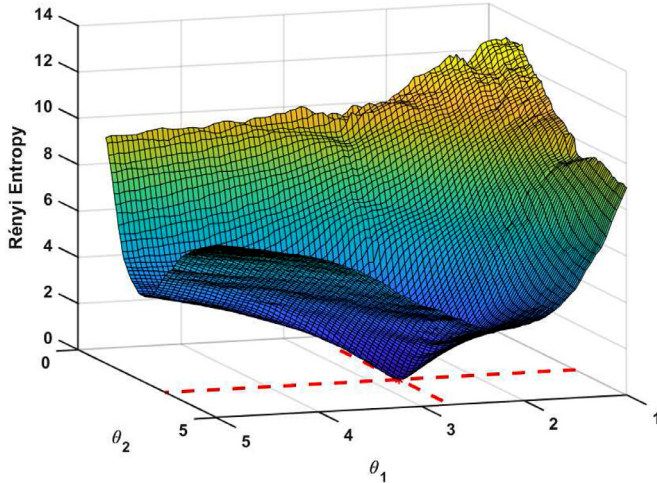


Fig. 10. The figure shows the Rényi entropy as a function of the proposal parameters of for a 2-D Gaussian function. The minimum matches the true parameter setting, marked by the crossing of the two dashed red lines. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

5.4. 2-D Transients

We go on to apply the estimator to the 2-D Gaussian defined in Section 3.6 and Eqs. (61) and (62) with $\theta = (e, \pi)$. We create a dictionary containing proposal parameters for both Gaussian components. For each pair of proposal parameters, we calculate the re-assignment and the subsequent Rényi surface. The resulting Rényi surface is depicted in Fig. 10. The true parameter values are shown with dashed red lines and the minimum of the Rényi surface aligns with those true parameter values.

6. Classification

In this section we use the parameter estimation algorithm as a classifier to identify the envelope, $a(t, \theta)$, of a signal

$$y(t, a) = \frac{1}{\sqrt{e_y}} (a(t - t_0, \theta) e^{i(\omega_0 t + \phi)} + z(t)) \quad (79)$$

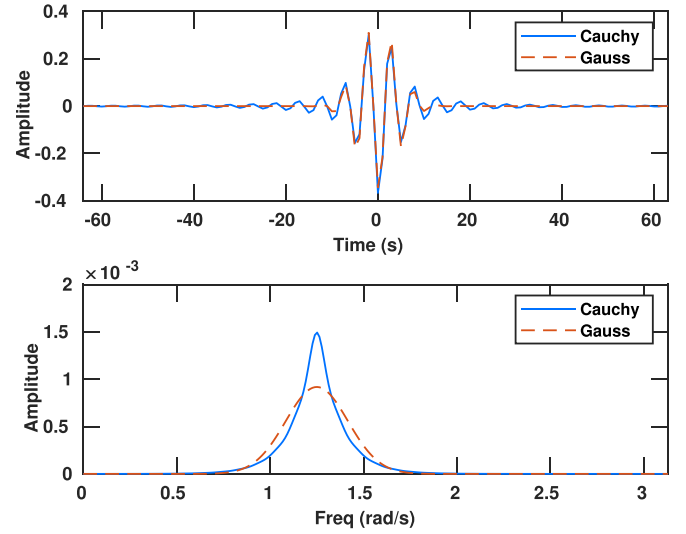


Fig. 11. The time and frequency representations of two noise-less, time, frequency and phase shifted signals, one with Cauchy signal function and the other with Gaussian signal function. Both signals have unit energy.

where the time shift is $t_0 \in \mathcal{U}(-64 + \delta_t, 64 - \delta_t)$, frequency shift is $\omega_0 \in \mathcal{U}(\delta_\omega, \pi - \delta_\omega)$, phase shift is $\phi \in \mathcal{U}(0, 2\pi)$, $\theta = 3\sqrt{2}$, $z(t) \in \mathcal{N}(0, \sigma)$ is stationary noise and $1/e_y$ is a constant such that the signal $y(t, a)$ has unit energy. The envelopes evaluated in this section are the Cauchy function, $a_C(\theta_C)$ from Eq. (56), and the Gaussian function, $a_G(\theta_G)$ from Eq. (77), because of their similar time and frequency representations, as demonstrated in Fig. 11. The δ_t and δ_ω are chosen so that at least 97.5% of the signal function is within the measurement time $[-64, 64]$ and frequency band $[0, \pi]$, with $\theta = 3\sqrt{2}$ thus making $\delta_t = 17$ (determined by the Cauchy function) and $\delta_\omega = 0.31$ (determined by the Gaussian function). Note that signal energy e_y is easily calculated for any measured signal and the normalization is needed to get accurate classification results.

The classification is binary, the result being either a Cauchy or Gaussian function, and the class is determined by finding the matching analysis window, i.e., the window that gives the lowest Rényi entropy. This means that for any given signal $y(t, a)$ with unknown $a(t, \theta)$, Algorithm 2 is applied with two different windows and sets of parameters, the unit energy Cauchy function, $h_C(q_C)$, and the unit energy Gaussian function, $h_G(q_G)$. The algorithm will for each window give an optimal parameter and the corresponding (minimum) Rényi entropy, thus the prediction for this MWR classification (MWRC) is the window function that solves the minimization problem

$$\hat{a}^{MWR}(t, q) = \arg \min_{h_C(q_C), h_G(q_G)} \left(\min_{q_C} R\{S_y^{h_C(q_C)}\}, \min_{q_G} R\{S_y^{h_G(q_G)}\} \right) \quad (80)$$

We evaluate the MWRC by comparing its performance to a similar approach using the spectrogram instead of the re-assigned spectrogram, the matched window spectrogram classification (MWSC), with prediction

$$\hat{a}^{MWS}(t, q) = \arg \min_{h_C(q_C), h_G(q_G)} \left(\min_{q_C} R\{S_y^{h_C(q_C)}\}, \min_{q_G} R\{S_y^{h_G(q_G)}\} \right) \quad (81)$$

and a Least Squares Envelope Classification (LSEC),

$$\hat{a}^{LSE}(t, q) = \arg \min_{h_C(q_C), h_G(q_G)} \left(\min_{q_C} \int_t (|y(t - \tilde{t}_0, a)| - h_C(q_C))^2 dt, \min_{q_G} \int_t (|y(t - \tilde{t}_0, a)| - h_G(q_G))^2 dt \right) \quad (82)$$

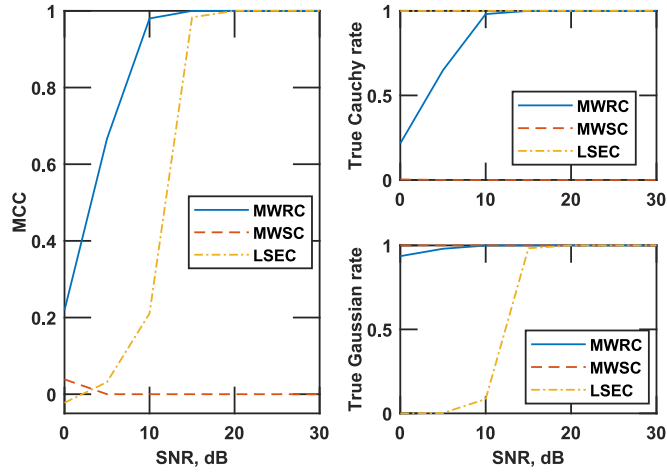


Fig. 12. Matthews correlation coefficient, true Cauchy rate and true Gaussian rate for MWRC, MWSC and LSEC on simulated signals disturbed by Gaussian noise.

where the time center is estimated as $\tilde{t}_0 = \arg \max_t |y(t, a)|$. All three classification methods are tested on the same simulated data consisting of a set of signals $y(t, a_C)$ and $y(t, a_G)$, each set having the same time, frequency and phase shift, but independent noise. The variance of the noise is adjusted so that signal sets with SNR_{dB} levels 0, 5, 10, ..., 30 dB are obtained. The classification is done on 1000 simulations of each signal set and SNR_{dB} , with different noise realizations, time, frequency and phase shifts, this yields the results presented in Fig. 12. The left plot shows the Matthews correlation coefficient (MCC) [37], which can be interpreted as a correlation coefficient between the observed and predicted classification, where +1 is perfect prediction, 0 no better than random prediction and -1 total disagreement between prediction and observation. The two right plots show the true Cauchy and true Gaussian rates (prediction and observation matching), indicating if any classification method favours one prediction over the other.

The MCC shows that the MWRC works very well for $SNR_{dB} \geq 10$, decently for $SNR_{dB} 5$, and outperforms the other classifiers. The true Cauchy and Gaussian rates show that the MWRC favours the Gaussian function when SNR_{dB} is low and the LSEC favours the Cauchy function for low and moderate SNR_{dB} , explained by the heavier tail in the time representation of the Cauchy function. The MWSC (almost) always predicts the Gaussian function, regardless of SNR_{dB} , and is thus no better than chance. Convergence of the MCC and true rates within 1000 simulations is verified by Fig. 13.

7. Real data application

We will finally apply the parameter estimator to two real data examples, one in 1-D and one in 2-D.

7.1. Parameter estimation of dolphin clicks

A dolphins click is commonly modeled as a Gaussian function, but this model does not always hold. In many cases the amplitude of the click is prolonged towards the end, giving the envelope of the signal a larger tail. Such an example of a recorded beluga whale (*Delphinapterus leucas*) is depicted in Fig. 14(a). We therefore define a Gumbel like model

$$y(t, \theta) = e^{-\frac{t}{\theta}} - e^{-\frac{t}{\theta}} \quad (83)$$

and use this as well as a Gaussian model, as candidates for classification of the envelope function. Following our proposed Algorithm 2, we estimate the scale parameter for the Gumbel model and Gaussian model separately.

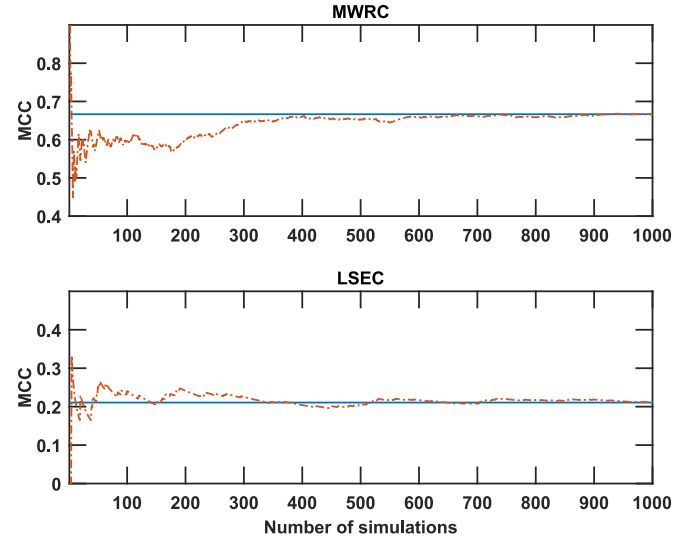


Fig. 13. Convergence of Matthews correlation coefficient for MWRC, $SNR_{dB} 5$ dB, and LSEC, $SNR_{dB} 10$ dB.

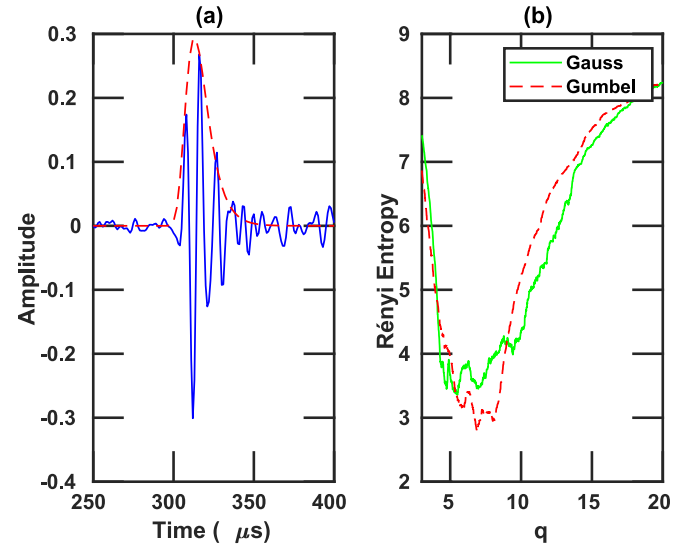


Fig. 14. Figure (a) depicts a recorded echo-location click from a dolphin. The dashed red line represents the estimated Gumbel envelope using the proposed parameter estimator. Figure (b) shows the Rényi curves for a Gaussian and Gumbel model, respectively. Note that the Gumbel model generates a lower entropy. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

The two resulting Rényi curves are depicted in Fig. 14(b). The best Gumbel model, $\theta = 6.88$, will give a reassigned spectrogram with a Rényi entropy of 2.79, this is lower than the minimum obtained by the Gaussian model, which is 3.36. Thus, the Gumbel model generate a better fit for this click. The estimated Gumbel envelope is depicted as a dashed red line in Fig. 14(a). The prolonged amplitudes towards the end of the click are well modeled by the estimated Gumbel envelope.

7.2. Parameter estimation of geological data

Seismograms are generated by an array of geophones measuring the vibrations of the soil as a function of time. The generated seismogram is hence 2-D and one can investigate how the

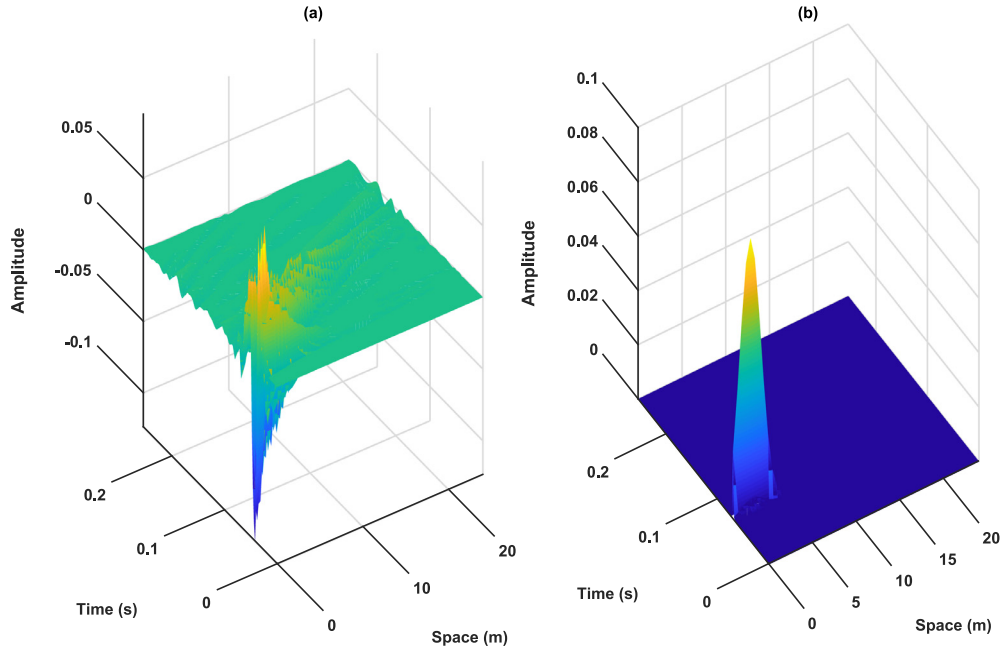


Fig. 15. Figure (a) depicts the recorded seismogram. Figure (b) depicts the envelope of the recorded seismogram, estimated using the proposed method. Note that the sinusoids propagating through the soil, in both time and space are not modeled here, only the impulse generated by the hammer strike.

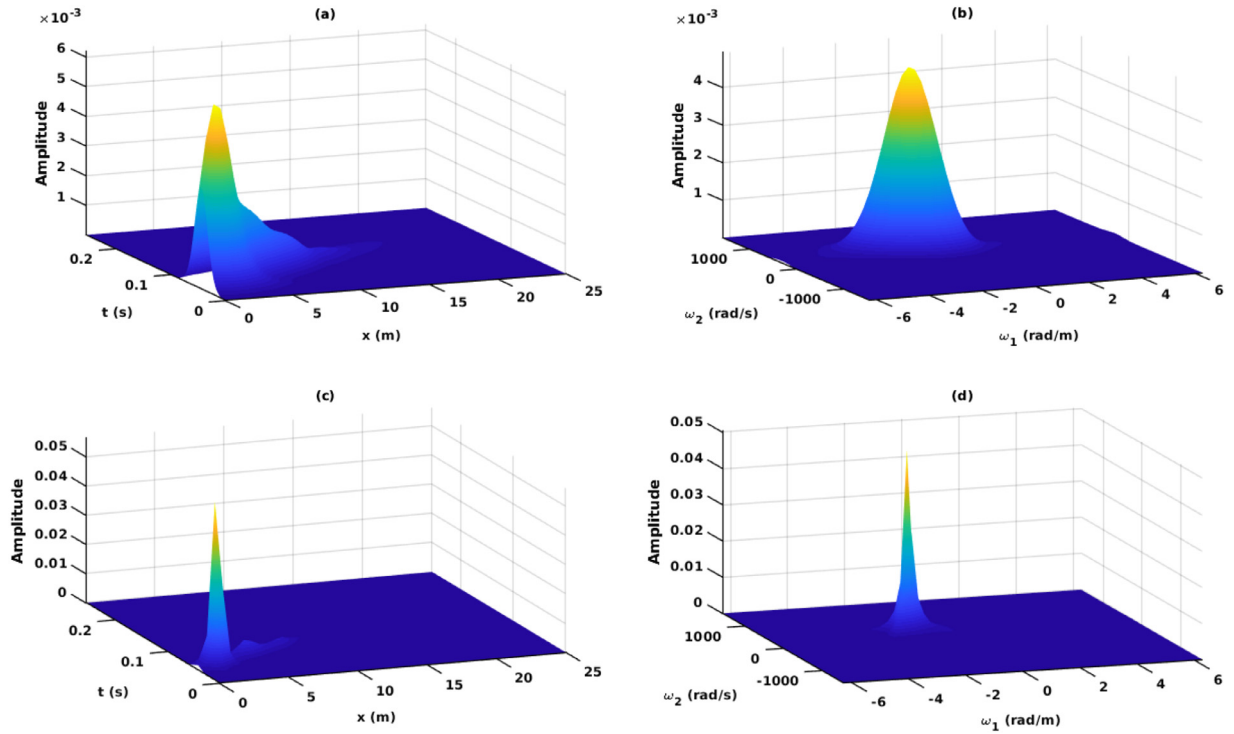


Fig. 16. The figure shows the marginals of the spectrogram and the MWR spectrogram of the 2-D seismogram example in Section 7.2. Figures (a) and (b) show the time and frequency marginal of the spectrogram, respectively. Figures (c) and (d) show the time and frequency marginal of the MWR spectrogram, respectively. Note that most energy has been reassigned to one single point, corresponding to the time and frequency shift of the function.

vibrations propagate in the soil both in time and space. An example of a seismogram is depicted in Fig. 15(a), where 49 receivers are equidistantly lined up 0.5 m apart. An active source, in this case hammer, is used to trigger the vibrations. The impulse response of the hammer may be modeled by a chirped 2-D Gaussian function. The Gaussian response is induced by the impulse of the hammer strike, and is then impeded in both time and space, by the soil. Note that the resulting sinusoidal waves propagating through the soil in both time and space will not be modeled here,

as they are not transient. The model used will be

$$y(t, r, \theta) = \exp\left(-\frac{1}{2(1-\theta_3^2)}\left[\frac{t}{\theta_1^2} + \frac{r}{\theta_2^2} - \frac{2\theta_3 tr}{\theta_1\theta_2}\right]\right) \quad (84)$$

$$\ast \exp\left(-i\frac{\theta_4 t^2}{2} - i\frac{\theta_5 r^2}{2}\right) \quad (85)$$

where θ_1 and θ_2 are the Gaussian shape parameters and θ_4 and θ_5 are the chirp rates in the two dimensions, respectively, whereas

θ_3 represents the correlation of the vibrations in time and space. The resulting estimated envelope is depicted in Fig. 15(b). The estimated shape parameters were $\theta_1 = 8.18 \cdot 10^{-4}$ and $\theta_2 = 1.20$, the time-space correlation was estimated as $\theta_3 = 3.44 \cdot 10^{-6}$ and the chirp rates to $\theta_4 = -6.94 \cdot 10^{-6}$ and $\theta_5 = 6.94 \cdot 10^{-3}$. The resulting marginals of the spectrogram and the MWR spectrogram are depicted in Fig. 16.

8. Conclusions

In this paper we have presented the matched window reassignment for n -dimensional signals, showing that any twice differentiable function used as the envelope shape of a transient oscillating signal may be reassigned to a single point in the time-frequency domain. We present a parameter estimator and give empirical evidence for its performance by showing a number of examples of different functions and comparing with existing methods and the Cramér-Rao Lower bound. A classifier is also derived that successfully separates similar signal envelope shapes of different functions and unknown parameter settings. The matched window reassignment gives reliable parameter estimates and classification performance even for signals with low SNR_{dB} levels.

Declaration of Competing Interest

The authors declare that they have no affiliations with or involvement in any organization or entity with any financial interest (such as honoraria; educational grants; participation in speakers' bureaus; membership, employment, consultancies, stock ownership, or other equity interest; and expert testimony or patent-licensing arrangements), or non-financial interest (such as personal relationships, affiliations, knowledge or beliefs) in the subject matter or materials discussed in this manuscript.

CRediT authorship contribution statement

Johan Brynolfsson: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing - original draft, Visualization, Project administration. **Isabella Reinhold:** Software, Validation, Formal analysis, Investigation, Data curation, Writing - review & editing, Visualization. **Maria Sandsten:** Conceptualization, Resources, Writing - review & editing, Supervision, Funding acquisition.

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